

Nuclear Medicine Notes

Radioactive Activity

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1 Radioactive Activity

Radioactive activity describes how quickly radioactive nuclei are undergoing nuclear transformations.

1.1 Activity Formula

Activity is defined as

$$A = -\frac{dN}{dt}$$

where

- A = radioactive activity
- N = number of radioactive nuclei
- t = time

The negative sign is present because the number of radioactive nuclei decreases as radioactive decay occurs.

Activity can also be expressed as

$$A = \lambda N$$

where

λ = decay constant.

Therefore,

$$A = -\frac{dN}{dt} = \lambda N$$

2 Decay Constant

The decay constant is represented by the Greek letter

λ .

In radioactive decay, λ represents the **decay constant**.

It should not be confused with the use of λ for wavelength in wave physics.

The decay constant is related to physical half-life by

$$\lambda = \frac{\ln 2}{T_{1/2}}$$

Since

$$\ln 2 \approx 0.693,$$

this can also be written as

$$\lambda = \frac{0.693}{T_{1/2}}$$

2.1 Units of the Decay Constant

The units of λ depend on the time unit used.

If time is measured in seconds,

$$\lambda \text{ has units of } \text{s}^{-1}.$$

If time is measured in hours,

$$\lambda \text{ has units of } \text{h}^{-1}.$$

For example,

$$\text{s}^{-1} = \frac{1}{\text{s}}$$

means “per second.”

Similarly,

$$\text{h}^{-1} = \frac{1}{\text{h}}$$

means “per hour.”

3 Physical Half-Life

Physical half-life is the time required for one-half of the radioactive nuclei in a sample to decay.

It is represented by

$$T_{1/2}.$$

The relationship between physical half-life and the decay constant is

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

or

$$T_{1/2} = \frac{0.693}{\lambda}$$

3.1 Relationship with Decay Constant

Physical half-life and decay constant are inversely related.

$$\boxed{T_{1/2} \uparrow \implies \lambda \downarrow}$$

and

$$\boxed{T_{1/2} \downarrow \implies \lambda \uparrow}$$

A radionuclide with a **short physical half-life** has a relatively large decay constant.

A radionuclide with a **long physical half-life** has a relatively small decay constant.

4 Technetium-99m Example

Technetium-99m has a physical half-life of approximately

$$T_{1/2} = 6 \text{ hours.}$$

Suppose a Tc-99m sample contains

$$N = 1.0 \times 10^{15}$$

radioactive nuclei.

We want to determine the activity of the sample.

4.1 Step 1: Convert Hours to Seconds

There are

$$3600 \text{ s}$$

in one hour.

Therefore,

$$6 \text{ h} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) = 21600 \text{ s.}$$

Thus,

$$\boxed{T_{1/2} = 21600 \text{ s}}$$

4.2 Step 2: Calculate the Decay Constant

Use

$$\lambda = \frac{\ln 2}{T_{1/2}}.$$

Substitute the half-life:

$$\lambda = \frac{0.693}{21600 \text{ s}}.$$

Therefore,

$$\boxed{\lambda = 3.21 \times 10^{-5} \text{ s}^{-1}}$$

4.3 Step 3: Calculate Activity

Use

$$A = \lambda N.$$

Substitute the values:

$$A = (3.21 \times 10^{-5} \text{ s}^{-1}) (1.0 \times 10^{15}).$$

When multiplying powers of ten,

$$10^{-5} \times 10^{15} = 10^{10}.$$

Therefore,

$$\boxed{A = 3.21 \times 10^{10} \text{ Bq}}$$

This is

$$32,100,000,000 \text{ Bq}.$$

In other words, approximately

$$\boxed{32.1 \text{ billion disintegrations occur every second}}$$

in the sample.

5 Becquerel

The Becquerel, abbreviated as

Bq,

is the SI unit of radioactive activity.

By definition,

$$1 \text{ Bq} = 1 \text{ disintegration per second}$$

or

$$1 \text{ Bq} = 1 \text{ s}^{-1}$$

when referring to radioactive transformations.

Therefore,

$$3.21 \times 10^{10} \text{ Bq}$$

means approximately

$$3.21 \times 10^{10}$$

radioactive disintegrations occur each second.

6 Disintegrations per Minute

DPM stands for

$$\text{Disintegrations Per Minute}$$

It expresses radioactive activity in terms of the number of nuclear disintegrations occurring every minute.

Since

$$1 \text{ minute} = 60 \text{ seconds},$$

and

$$1 \text{ Bq} = 1 \text{ disintegration per second},$$

then

$$1 \text{ Bq} = 60 \text{ dpm}$$

For example,

$$100 \text{ Bq} \times 60 = 6000 \text{ dpm}.$$

Therefore,

$$100 \text{ Bq} = 6000 \text{ dpm}$$

7 Curie and Radioactivity

7.1 Marie Curie

Marie Curie is associated with the discovery of the radioactive elements **radium** and **polonium**.

She also coined the term

$$\text{radioactivity}$$

to describe the phenomenon of spontaneous radiation emission from unstable atoms.

7.2 Radium and Polonium

Two important elements associated with the work of Marie Curie are

- **Radium**
- **Polonium**

These discoveries became important to the early understanding of radioactivity.

7.3 The Curie

The Curie is a unit of radioactive activity and is abbreviated as

Ci.

One Curie corresponds to

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ disintegrations per second}$$

Since

$$1 \text{ Bq} = 1 \text{ disintegration per second},$$

we can write

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ Bq}$$

The Curie was historically based on the activity associated with approximately

$$1 \text{ g}$$

of radium-226.

8 Amount Remaining After Multiple Half-Lives

After every physical half-life, one-half of the radioactive nuclei remain.

The amount remaining can be calculated using

$$N = \frac{N_0}{2^n}$$

where

- N_0 = original number of radioactive nuclei
- N = number of radioactive nuclei remaining
- n = number of half-lives that have passed

The number of half-lives can be calculated using

$$n = \frac{t}{T_{1/2}}$$

8.1 Example

Suppose

$$N_0 = 800,$$

the physical half-life is

$$T_{1/2} = 6 \text{ hours},$$

and the total elapsed time is

$$t = 18 \text{ hours}.$$

First determine the number of half-lives:

$$n = \frac{18}{6} = 3.$$

Therefore,

$$N = \frac{800}{2^3}.$$

Since

$$2^3 = 8,$$

we obtain

$$N = \frac{800}{8} = \boxed{100}.$$

This can also be visualized as

$$800 \xrightarrow{6\text{ h}} 400 \xrightarrow{6\text{ h}} 200 \xrightarrow{6\text{ h}} 100.$$

The decay does not stop at 100.

If another half-life passes,

$$100 \rightarrow 50,$$

and another gives

$$50 \rightarrow 25.$$

9 Key Formulas

9.1 Activity

$$A = -\frac{dN}{dt}$$

$$A = \lambda N$$

9.2 Decay Constant

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{T_{1/2}}$$

9.3 Physical Half-Life

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

9.4 Number Remaining

$$N = \frac{N_0}{2^n}$$

with

$$n = \frac{t}{T_{1/2}}$$

9.5 Activity Units

$$1 \text{ Bq} = 1 \text{ disintegration per second}$$

$$1 \text{ Bq} = 60 \text{ dpm}$$

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ Bq}$$